FW Test Report

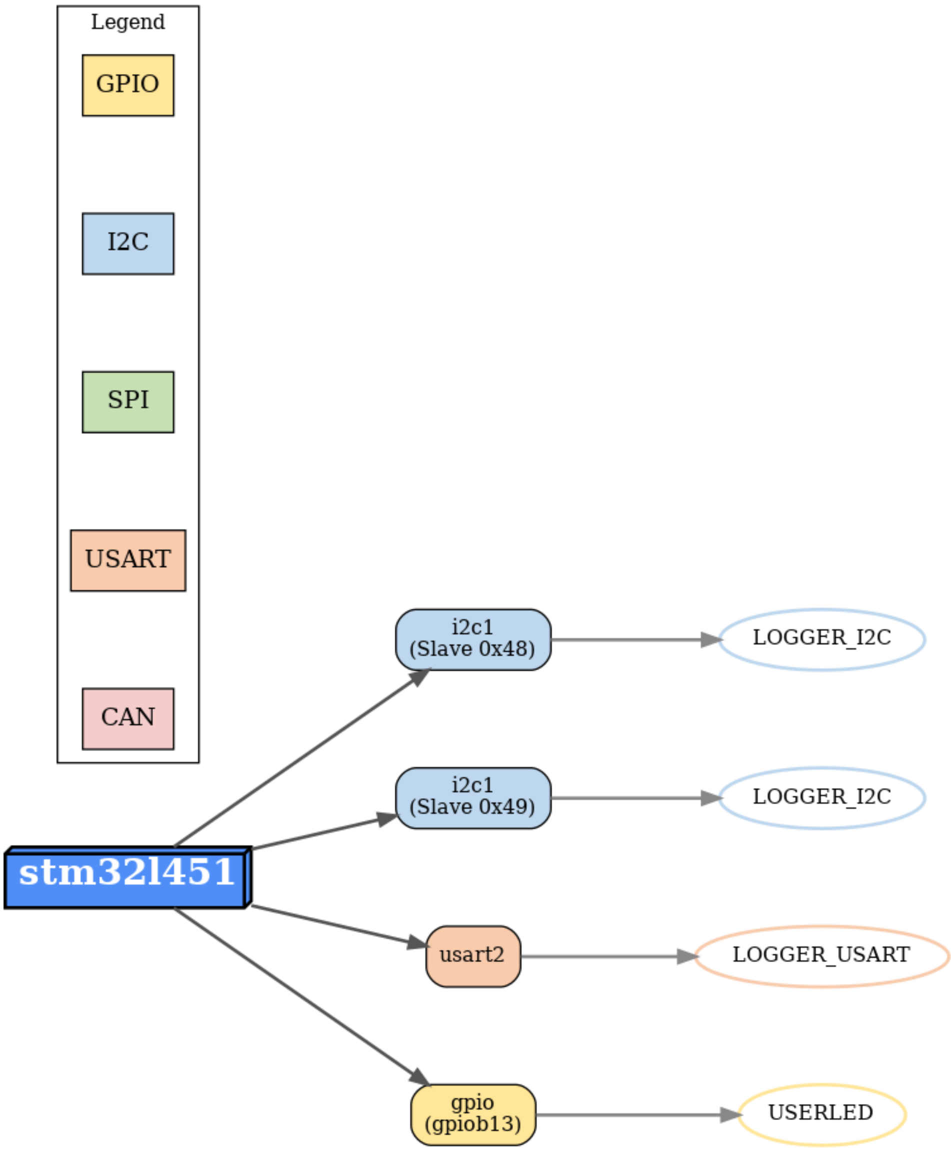
Generated on: 2026-06-28 16:54:30

MCU: **stm32l451**

Tested Peripherals

#	Peripheral	Sensor	Port	Pin
1	i2c1	LOGGER_I2C		
2	i2c1	LOGGER_I2C		
3	usart2	LOGGER_USART		
4	gpio	USERLED	gpiob	13

System Diagram



Logs

16:54:04.8624 **[INFO]** Including script: /workspace/uploads/example.resc
16:54:04.8770 **[INFO]** System bus created.
16:54:05.5528 **[WARNING]** Lptimer not passed in the RCC constructor. Changes to the low-power timer clock will be ignored
16:54:06.1045 **[INFO]** sysbus: Loaded SVD: /workspace/svds/stm32l451.svd. Name: STM32L451. Description: STM32L451.
16:54:06.7045 **[INFO]** machine-0: CPUs: ["machine-0.cpu"] were added to a new GDB server created on port :3333
16:54:06.7046 **[INFO]** machine-0: GDB server with all CPUs started on port :3333
16:54:06.9223 **[INFO]** sysbus: Loading block of 89768 bytes length at 0x8000000.
16:54:06.9367 **[INFO]** sysbus: Loading block of 30424 bytes length at 0x8015EA8.
16:54:06.9367 **[INFO]** sysbus: Loading block of 1536 bytes length at 0x8016558.
16:54:07.3109 **[INFO]** cpu: Guessing VectorTableOffset value to be 0x8000000.
16:54:07.3143 **[INFO]** cpu: Setting initial values: PC = 0x80016F9, SP = 0x20020000.
16:54:07.3164 **[INFO]** machine-0: Machine started.
16:54:07.4021 **[WARNING]** sysbus: [cpu: 0x800C89A] (tag: 'PWR_CR1') ReadDoubleWord from non existing peripheral at 0x40007000, returning 0x00000F00.
16:54:07.4021 **[WARNING]** sysbus: [cpu: 0x800C8A8] (tag: 'PWR_CR1') ReadDoubleWord from non existing peripheral at 0x40007000, returning 0x00000F00.
16:54:07.4032 **[WARNING]** sysbus: [cpu: 0x800C8B4] WriteDoubleWord of value 0xB00 to an unimplemented register PWR:CR1 (0x40007000) generated from SVD
16:54:07.4035 **[WARNING]** sysbus: [cpu: 0x800C8D6] ReadDoubleWord from an unimplemented register PWR:SR2 (0x40007014), returning a value from SVD: 0x0
16:54:07.4035 **[WARNING]** sysbus: [cpu: 0x800C8EA] ReadDoubleWord from an unimplemented register PWR:SR2 (0x40007014), returning a value from SVD: 0x0
16:54:07.4035 **[WARNING]** sysbus: [cpu: 0x800C842] (tag: 'PWR_CR1') ReadDoubleWord from non existing peripheral at 0x40007000, returning 0x00000F00.
16:54:07.4036 **[WARNING]** sysbus: [cpu: 0x800C84A] WriteDoubleWord of value 0xF00 to an unimplemented register PWR:CR1 (0x40007000) generated from SVD
16:54:07.4050 **[WARNING]** sysbus: [cpu: 0x8009710] (tag: 'PWR_CR1') ReadDoubleWord from non existing peripheral at 0x40007000, returning 0x00000F00.
16:54:07.4150 **[WARNING]** sysbus: [cpu: 0x800B630] ReadDoubleWord from an unimplemented register SYSCFG:EXTICR4 (0x40010014), returning a value from SVD: 0x0
16:54:07.4150 **[WARNING]** sysbus: [cpu: 0x800B6C4] WriteDoubleWord of value 0x1 to an unimplemented register SYSCFG:EXTICR4 (0x40010014) generated from SVD
16:54:07.4151 **[WARNING]** sysbus: [cpu: 0x800B6CA] ReadDoubleWord from an unimplemented register EXTI:RTSR1 (0x40010408), returning a value from SVD: 0x0
16:54:07.4151 **[WARNING]** sysbus: [cpu: 0x800B6F0] WriteDoubleWord of value 0x1000 to an unimplemented register EXTI:RTSR1 (0x40010408) generated from SVD
16:54:07.4151 **[WARNING]** sysbus: [cpu: 0x800B6F4] ReadDoubleWord from an unimplemented register EXTI:FTSR1 (0x4001040C), returning a value from SVD: 0x0
16:54:07.4152 **[WARNING]** sysbus: [cpu: 0x800B71A] WriteDoubleWord of value 0x1000 to an unimplemented register EXTI:FTSR1 (0x4001040C) generated from SVD
16:54:07.4153 **[WARNING]** sysbus: [cpu: 0x800B71E] ReadDoubleWord from an unimplemented register EXTI:EMR1 (0x40010404), returning a value from SVD: 0x0
16:54:07.4153 **[WARNING]** sysbus: [cpu: 0x800B744] WriteDoubleWord of value 0x0 to an unimplemented register EXTI:EMR1 (0x40010404) generated from SVD
16:54:07.4153 **[WARNING]** sysbus: [cpu: 0x800B748] ReadDoubleWord from an unimplemented register EXTI:IMR1 (0x40010400), returning a value from SVD: 0xFF820000
16:54:07.4154 **[WARNING]** sysbus: [cpu: 0x800B76E] WriteDoubleWord of value 0xFF821000 to an unimplemented register EXTI:IMR1 (0x40010400) generated from SVD
16:54:07.4177 **[WARNING]** sysbus: [cpu: 0x800A306] (tag: 'PWR_CR1') ReadDoubleWord from non existing peripheral at 0x40007000, returning 0x00000F00.
16:54:07.4177 **[WARNING]** sysbus: [cpu: 0x800A30E] WriteDoubleWord of value 0xF00 to an unimplemented register PWR:CR1 (0x40007000) generated from SVD
16:54:07.4177 **[WARNING]** sysbus: [cpu: 0x800A32E] (tag: 'PWR_CR1') ReadDoubleWord from non existing peripheral at 0x40007000, returning 0x00000F00.
16:54:07.4247 **[WARNING]** gpiob: Unhandled write to offset 0x4. Unhandled bits: [8] when writing value 0x100. Tags: OT8 (0x1).
16:54:07.4254 **[WARNING]** gpiob: Unhandled write to offset 0x4. Unhandled bits: [9] when writing value 0x200. Tags: OT9 (0x1).

16:54:07.4263 **[WARNING]** i2c1: Unhandled write to offset 0x10. Unhandled bits: [0, 7-8, 11, 13, 16, 20-23] when writing value 0xF12981. Tags: SCLL (0x81), SCLH (0x29), SDADEL (0x1), SCLDEL (0xF).
16:54:07.4266 **[INFO]** i2c1: Slave address 1: 0x0, mode: 7-bit, status: disabled
16:54:07.4266 **[INFO]** i2c1: Slave address 1: 0x0, mode: 7-bit, status: enabled
16:54:07.4275 **[WARNING]** i2c1: Unhandled write to offset 0x4. Unhandled bits: [15] when writing value 0x2008000. Tags: NACK (0x1).
16:54:07.4276 **[INFO]** i2c1: Slave address 2: 0x0, mask: 0x0, status: disabled
16:54:07.4286 **[INFO]** i2c1: Slave address 2: 0x0, mask: 0x0, status: disabled
16:54:07.4318 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [14] when writing value 0x4000. Tags: DEM (0x1).
16:54:07.4319 **[WARNING]** usart2: Unhandled write to offset 0x0. Unhandled bits: [16-17, 21-22] when writing value 0x63000C. Tags: DEDT (0x3), DEAT (0x3).
16:54:07.4325 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:07.4326 **[WARNING]** usart1: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:07.4544 **[NOISY]** i2c1.LOGGER_I2C1: Write 0
16:54:07.4545 **[NOISY]** i2c1.LOGGER_I2C1: Read 1 number of bytes
16:54:07.4838 **[WARNING]** nvic: Trying to set the priority for interrupt 16 to 0xFF, but it should be maskable with 0xF0
16:54:12.3205 **[NOISY]** i2c1.LOGGER_I2C2: Write 1
16:54:12.3208 **[NOISY]** i2c1.LOGGER_I2C2: Write D5 83
16:54:12.3210 **[NOISY]** i2c1.LOGGER_I2C2: Write 0
16:54:12.3210 **[NOISY]** i2c1.LOGGER_I2C2: Read 2 number of bytes
16:54:12.3216 **[NOISY]** i2c1.LOGGER_I2C2: Write 1
16:54:12.3216 **[NOISY]** i2c1.LOGGER_I2C2: Write D5 83
16:54:12.3216 **[NOISY]** i2c1.LOGGER_I2C2: Write 0
16:54:12.3216 **[NOISY]** i2c1.LOGGER_I2C2: Read 2 number of bytes
16:54:12.3217 **[NOISY]** i2c1.LOGGER_I2C2: Write 1
16:54:12.3217 **[NOISY]** i2c1.LOGGER_I2C2: Write C5 83
16:54:12.3218 **[NOISY]** i2c1.LOGGER_I2C2: Write 0
16:54:12.3218 **[NOISY]** i2c1.LOGGER_I2C2: Read 2 number of bytes
16:54:13.5558 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:13.5559 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:13.5559 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:13.5560 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:13.5560 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:13.5560 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:13.5561 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:13.5564 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:15.5567 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:15.5568 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:15.5568 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:15.5569 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:15.5570 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:15.5570 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:15.5576 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:15.5577 **[WARNING]** usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:15.8210 **[NOISY]** gpiob.USERLED1: LED state changed to True

16:54:17.3196 [NOISY] i2c1.LOGGER_I2C2: Write 1
16:54:17.3196 [NOISY] i2c1.LOGGER_I2C2: Write D5 83
16:54:17.3196 [NOISY] i2c1.LOGGER_I2C2: Write 0
16:54:17.3196 [NOISY] i2c1.LOGGER_I2C2: Read 2 number of bytes
16:54:17.3196 [NOISY] i2c1.LOGGER_I2C2: Write 1
16:54:17.3196 [NOISY] i2c1.LOGGER_I2C2: Write D5 83
16:54:17.3197 [NOISY] i2c1.LOGGER_I2C2: Write 0
16:54:17.3197 [NOISY] i2c1.LOGGER_I2C2: Read 2 number of bytes
16:54:17.3207 [NOISY] i2c1.LOGGER_I2C2: Write 1
16:54:17.3207 [NOISY] i2c1.LOGGER_I2C2: Write C5 83
16:54:17.3207 [NOISY] i2c1.LOGGER_I2C2: Write 0
16:54:17.3207 [NOISY] i2c1.LOGGER_I2C2: Read 2 number of bytes
16:54:17.5625 [WARNING] usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:17.5626 [WARNING] usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:17.5626 [WARNING] usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:17.5626 [WARNING] usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:17.5626 [WARNING] usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:17.5627 [WARNING] usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:17.5627 [WARNING] usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:17.5627 [WARNING] usart2: Unhandled write to offset 0x8. Unhandled bits: [0] when writing value 0x1. Tags: EIE (0x1).
16:54:17.8216 [NOISY] gpiob.USERLED1: LED state changed to False
16:54:22.3195 [NOISY] i2c1.LOGGER_I2C2: Write 1
16:54:22.3196 [NOISY] i2c1.LOGGER_I2C2: Write D5 83
16:54:22.3196 [NOISY] i2c1.LOGGER_I2C2: Write 0
16:54:22.3196 [NOISY] i2c1.LOGGER_I2C2: Read 2 number of bytes
16:54:22.3196 [NOISY] i2c1.LOGGER_I2C2: Write 1
16:54:22.3196 [NOISY] i2c1.LOGGER_I2C2: Write D5 83
16:54:22.3197 [NOISY] i2c1.LOGGER_I2C2: Write 0
16:54:22.3197 [NOISY] i2c1.LOGGER_I2C2: Read 2 number of bytes
16:54:22.3203 [NOISY] i2c1.LOGGER_I2C2: Write 1
16:54:22.3204 [NOISY] i2c1.LOGGER_I2C2: Write C5 83
16:54:22.3204 [NOISY] i2c1.LOGGER_I2C2: Write 0
16:54:22.3204 [NOISY] i2c1.LOGGER_I2C2: Read 2 number of bytes

Customer Test Script

```python
import time
import socket
import threading

COMMANDS = {
    "SET_LOG_DEBUG": [0x01, 0x06, 0x04, 0x02, 0x00, 0x01, 0xE8, 0xFA],
    "SET_TPS_ENABLE_0": [0x01, 0x06, 0x04, 0x01, 0x00, 0x00, 0xD9, 0x3A],
    "SET_TPS_ENABLE_1": [0x01, 0x06, 0x04, 0x01, 0xFF, 0xFF, 0xD8, 0x8A]
}


class ModbusClient:
    def __init__(self, host="localhost", port=1234):
        self.sock = socket.create_connection((host, port))

        self.rx_thread = threading.Thread(
            target=self._rx_loop,
            daemon=True
        )
        self.rx_thread.start()

    def _rx_loop(self):
        buffer = ""

        while True:
            try:
                data = self.sock.recv(4096)

                if not data:
                    print("[RX] disconnected")
                    break

                buffer += data.decode("ascii", errors="replace")

                while "\n" in buffer:
                    line, buffer = buffer.split("\n", 1)

                    line = line.rstrip("\r")

                    if line:
                        print(f"[RX] {line}")

            except Exception as e:
                print("[RX ERROR]", e)
                break

    def send(self, cmd_name):
        if cmd_name not in COMMANDS:
            raise ValueError(f"Unknown command: {cmd_name}")

        payload = bytes(COMMANDS[cmd_name])
```

```
        print(
            "[TX]",
            cmd_name,
            ":",
            " ".join(f"{b:02X}" for b in payload)
        )

        self.sock.sendall(payload)

    def close(self):
        self.sock.close()


# Logic

mb = ModbusClient("localhost", 1234)

mb.send("SET_LOG_DEBUG")

time.sleep(2)
mb.send("SET_TPS_ENABLE_1")

time.sleep(2)
mb.send("SET_TPS_ENABLE_0")

time.sleep(20)

mb.close()
```

Customer Test Logs

```
[TX] SET_LOG_DEBUG : 01 06 04 02 00 01 E8 FA
[RX] [MDBS] data write by callback
[RX] data[0]: 0x1
[RX] [MDBS] Sending response: 01 06 04 02 00 01 E8 FA
[TX] SET_TPS_ENABLE_1 : 01 06 04 01 FF FF D8 8A
[RX] [MDBS] data write by callback
[RX] data[0]: 0xFFFF
[RX] [MDBS] Sending response: 01 06 04 01 FF FF D8 8A
[RX] Gas reading: 0.000000
[RX] Gas Raw reading: 0
[RX] Temperature reading: 192.009995
[TX] SET_TPS_ENABLE_0 : 01 06 04 01 00 00 D9 3A
[RX] [MDBS] data write by callback
[RX] data[0]: 0x0
[RX] [MDBS] Sending response: 01 06 04 01 00 00 D9 3A
[RX] Gas reading: 0.000000
[RX] Gas Raw reading: 0
[RX] Temperature reading: 192.009995
[RX] Gas reading: 0.000000
[RX] Gas Raw reading: 0
[RX] Temperature reading: 192.009995
```